UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT :

Network Design for Faculty of Computing Block N28c

TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

SECR1213 - NETWORK COMMUNICATIONS
SEMESTER I, SESSION 2024/2025
SECTION 02

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1.0 Introduction

In Task 2, the project progresses from the setup phase to the initial design stage, where the focus shifts to preliminary analysis. This phase involves understanding and designing the network requirements for the new N28c building of the Faculty of Computing. The analysis emphasizes evaluating the infrastructure, security measures, and unique technological needs of different areas within the building, such as labs, classrooms, and the student lounge. By addressing key questions and conducting in-depth research, the analysis delves into essential components, including network infrastructure, required bandwidth, data protection, and monitoring systems. This thorough assessment ensures that the network design is practical, reliable, and capable of meeting future demands. Ultimately, this preliminary review aims to confirm the project's technical feasibility and alignment with the Faculty’s goals for a secure, scalable, and high-performance network.

2.0 Questions

1. What are the reliable backup strategies for ensuring the protection, continuity, and integrity of data in server and database rooms?

The best strategy is where we use a hybrid backup strategy that leverages both on-premises storage and the public cloud as backup destinations, offering a balanced approach to data protection. By combining these two options, the faculty can achieve greater resilience and flexibility. The on-premises storage allows for fast recovery times, particularly for recent data, since local access avoids the latency associated with downloading data from the cloud. For example we can use devices like Network Attached Storage (NAS) or Storage Area Networks (SAN) located in the server room. This setup is especially beneficial for large files or applications that require minimal downtime. Meanwhile, the cloud serves as a secondary, secure location for data, adding a layer of protection against on-site disasters, such as hardware failure or physical damage. For example, Amazon Web Services (AWS) offers cloud backup solutions that allow users to store data securely in remote data centers with encryption and automated backup options, ensuring data safety and accessibility even if something happens to the main facility. In addition, using a hybrid approach can be cost-effective: frequently accessed or critical data remains on-premises, while older or less critical backups can be moved to the cloud, reducing long-term storage and retrieval costs.

2. How should we address scalability in the network design to accommodate future growth in student and staff numbers?

To accommodate future growth, network design should incorporate a few key strategies. First, a modular architecture allows for efficient expansion by segmenting the network into distinct layers, such as core, distribution, and access. This design makes it easier to add new sections or branches without disrupting the entire network . Cloud integration is another essential component, as it enables flexible resource allocation and scalability through hybrid or multi-cloud setups. By using the cloud, the network can handle variable loads and expand capacity as needed. Additionally, IPv6 adoption is critical to support the rising number of connected devices and avoid address exhaustion, as IPv6 provides a much larger address space than IPv4 . Finally, creating a network infrastructure that is both flexible and designed for growth is essential. This involves choosing hardware and systems that can be easily expanded or upgraded, such as modular routers and switches, so that the network can handle increasing traffic without major overhauls. By adding capacity as needed, the network stays reliable and efficient even as demand grows. These strategies together create a robust, adaptable network that can scale effectively over time.

3. What is the expected internet speed and bandwidth capacity needed to support high-speed connections in each area?

The expected internet speed and bandwidth capacity required to support high-speed connections vary depending on the area and the specific use case. For a hybrid classroom , video conferencing room or a lab with 5 to 50 users require internet speeds of **100 Mbps to 1 Gbps** to accommodate tasks like VoIP calls, video conferencing, and cloud services. A bandwidth capacity of **1 Gbps** is ideal for ensuring multiple users can access high-speed services simultaneously. Large enterprises like a student lounge with over 50 users, handling high-definition video conferencing, large file transfers, and real-time applications, require speeds of **1 Gbps or higher**, with a bandwidth capacity of **1 Gbps to 10 Gbps** to avoid network slowdowns. For data centers and cloud infrastructure, internet connections typically range from **10 Gbps to 100 Gbps** to manage large-scale data flows and high traffic volumes. In rural or remote locations with limited infrastructure, internet speeds can range from **10 Mbps to 100 Mbps**, although upgraded technologies like LTE or 5G may offer speeds between **100 Mbps and 1 Gbps** depending on coverage. Overall, the required speeds and bandwidth capacities vary widely, depending on the number of users and the intensity of the internet activities in each area.

4) Should network segmentation be implemented to isolate traffic across different areas, such as laboratories, classrooms, and the student lounge, for optimal security and performance?

Yes. For our plan, the network segmentation using **Virtual LANs (VLANs)** will be highly recommended to enhance the network security and performance. By segmenting the network, the network traffic in each area like the labs, classrooms, and student lounge will be isolated, which is able to **reduce the problem of network congestion**. Therefore, this can **maximize the use of bandwidth** in each area. Moreover, VLANs can provide a **security layer** to each dedicated lab environment like the Cisco Networking Lab(CNL) and Embedded System Lab(ESL) by **isolating the essential data** from unauthorized access. As a result, VLANs provide a **secure and manageable network infrastructure** which is able to support the unique needs for each facility in the building.

5) What type of cabling is most suitable for use within this network infrastructure, considering factors like speed, durability, and cost?

Since the cost is around RM2.5 million budget, we decided to use the combination of **Category 6a (Cat6a) twisted-pair** and **fiber optic cabling** for the building. The Cat6a is more cost-effective which is capable of supporting **10 Gbps speed**. Therefore, this type of cabling will be ideal for most internal connections within each lab, classroom and student lounge. While for the fiber optic cabling, it will be more expensive since it provides higher speed and longer-distance capability, thus it will be only used for critical areas like the Cisco Networking Lab(CNL) and Embedded System Lab(ESL) to ensure **high-speed** and **low-latency** of network performance. This combination of cabling will provide reliable connectivity, while controlling the cost and allow for future scalability.

6.) Do we need a backup power supply for our building to ensure 24-hour operation of the server room and database room?

Yes, for the new Faculty of Computing (FC) building, N28c having backup power is essential to ensure 24-hour operation of the server room and any associated infrastructure, such as the network Operations Center. This is important to **maintain continuous access** to high-performance Internet, **uninterrupted data services**, and key educational resources needed for teaching and research. With backup power systems, such as **uninterruptible power supplies (UPS) and generators**, the building protects against potential data loss, hardware damage, and downtime that can interrupt student and faculty activities. This ensures that services such as Cisco Networking LAB, Embedded Systems LAB, video conferencing, and hybrid classrooms operate even during outages. Implementing these backup systems will provide reliability, support academic continuity, and protect research and data integrity within FC facilities.

7.) What are the recommended computer specifications for each lab?

For the new Faculty of Computing (FC) building, computer specifications should be tailored to the specific needs of each laboratory. **General LABS** can use PCS equipped with **Intel Core i5 processors, 16gb RAM, and 512gb ssd** to efficiently handle programming, data analysis, and office tasks. **Cisco networking LAB** should be equipped with more powerful PCS, such as those with **Intel Core i7 processors and 16gb of RAM**, to run network simulation software smoothly. The Embedded Systems Lab requires a high-performance computer with at least an **Intel Core i7 processor, 32 GB of RAM, and 1 TB SSD** to support complex iot and embedded projects. All LABS should have a fast Internet connection (Gigabit Ethernet and Wi-Fi 6) and reliable backup power to prevent outages. This ensures that students and staff can complete advanced computing tasks without performance or connectivity issues.

8.) What type of network architecture (examples: p2p and CS) will be applied for each lab?

For the **General Purpose Labs**, a **Client-Server (CS) architecture** would be suitable as it provides a centralization of control and resource management. Furthermore, this setup is more focused on data security to ensure data stored on a server that is regularly backed up which can minimize the risks of data loss if any individual workstations happen to encounter unpredicted issues. A **Peer-to-Peer (P2P) network** alongside a **Client-Server (CS) architecture** may be beneficial here for the **Cisco Network Lab**. P2P enables students to practice direct device-to-device connections without centralized control by setting up simulated networks and experiencing real-world networking scenarios, while CS can manage access to resources and maintain administrative oversight. As for the **Embedded Lab**, a **Client-Server (CS) architecture** with an IoT gateway will suffice for managing the devices' connectivity securely. The CS model allows for controlled access and security, while an IoT gateway can handle the large variety of devices and protocols typical in an IoT setting.

9.) Do we need dedicated cooling systems for the server and network operation center rooms?

Yes, dedicated cooling systems are essential for the server and network operation center (NOC) rooms because the equipment used generates significant heat, which can lead to hardware malfunctions or failures if not managed properly. A dedicated cooling system will ensure optimal performance by keeping devices at a stable temperature to avoid overheating. By doing so will extend the equipment lifespan which prevent network disruptions caused by heat-related shutdowns. An independent cooling solution, such as a precision cooling system specifically designed for server rooms, would be optimal to maintain consistent temperatures. In addition the cooling system needs to be placed below the floor to avoid condensation of water vapors to form on the surface of the servers and this can also help temperature regulation of the servers.

10.) Should the data be stored locally on physical servers or in the cloud?

Both local physical servers and cloud storage have both their own pros and cons that depend on security, cost, and scalability needs. **Local Physical Servers** even offer direct control over data, with minimal third-party dependency which is suitable for highly secure or sensitive data, where regulatory compliance mandates data residency. In contrast, this leads to a higher upfront cost, physical maintenance requirements, and limited scalability. **Cloud Storage** on the other hand is scalable, flexible, and often more cost-effective. Cloud providers typically offer robust disaster recovery and backup solutions, and they are more convenient for remote access. However, its dependence on third-party providers raises potential privacy and security concerns. There can also be recurring costs and issues if internet access is disrupted. As a conclusion, a hybrid model might be ideal for this setup. Critical data can remain on local servers for security, while less sensitive data and applications could be hosted in the cloud for flexibility and ease of access. This allows FC to balance control with scalability, preparing for future needs without incurring unnecessary costs.

11.) What user authentication and access controls should be implemented to ensure computer and data security in general lab?

To protect each lab's computers and data, a multi-tiered approach should be employed, including multi-factor authentication (MFA) for enhanced login security, role-based access control (RBAC) to assign specific access rights based on user roles, and Network access control (NAC) solutions such as Cisco's Identity Services Engine (ISE). To authenticate and profile the device. Integrating these measures with a centralized directory service such as Microsoft Active directory simplifies the management of user credentials and permissions across all LABS, ensuring that only authorized personnel have access to sensitive resources.

12.) What specialized networking equipment does the Cisco Networking Lab need to ensure practical, hands-on learning?

In order to do actual hands-on learning in Cisco networking LABS, specialized networking equipment is essential. LABS should be equipped with routers such as the Cisco ISR 4000 series, Layer 3 switches such as the Cisco Catalyst 9000 series for advanced network configuration, and Cisco ASA or Fire firewalls for security training. In addition, Cisco Aironet or Meraki wireless access points should include wireless networking exercises, and software tools such as Cisco Packet Tracker and GNS3 can complement simulation hands-on training.

13.) Should the embedded Systems lab have a dedicated power circuit for specialized equipment? If so, how should these circuits be set up?

Embedded systems LABS should have dedicated power circuits to ensure stable power delivery and prevent electrical interference to sensitive devices such as microcontrollers and sensors. Installing isolated power outlets on separate circuits, supplemented by an uninterruptible power supply (UPS) with surge protection, is essential to protect equipment and maintain functionality during outages. An automated power distribution unit (pdu) with real-time monitoring and a grounded power bar should also be used to safely manage the distribution and meet electrical safety standards.

3.0 Feasibility Assessment

After gathering information by answering the provided questions, we can now assess the feasibility of the network project for the Faculty of Computing. The feasibility assessment involves evaluating the technical, economic, and operational aspects of the project to determine whether it is viable.

Operational Feasibility: For the new Faculty of Computing building N28c, the proposed solution is operationally feasible and well aligned with the goals of reliability, scalability, and data protection. A hybrid backup strategy utilizing local storage (NAS/SAN) and cloud services ensures fast local data recovery while providing additional disaster recovery and resilience layers through cloud storage. To accommodate future growth, an IPv6 modular network design can be seamlessly expanded and supports more connected devices. VLAN network segmentation across laboratories, classrooms, and student lounges enhances security and performance by isolating traffic and reducing congestion. The combination of Cat6a and fiber cabling balances cost and performance, providing high-speed connectivity throughout the building, and fiber ensures low latency in critical areas. Dedicated backup power and precision cooling systems are used in server and network operations center rooms to prevent downtime, hardware damage, and overheating, ensuring 24/7 operation. Customized computer specifications meet the unique needs of each laboratory, ranging from high-performance PCs for Cisco networking labs to more standard setups for general laboratories. The use of a hybrid data storage approach - keeping critical data on local servers while utilizing cloud storage for less sensitive data - provides a balance between control, scalability, and cost-effectiveness. These strategies collectively form a powerful, future-oriented infrastructure that supports FC's academic and operational needs, ensuring reliable, high-performance services and adaptability to future requirements.

Economic Feasibility: Considering the budget that has been provided with a total of RM 2.5 million for this project.To ensure financial sustainability and viability, careful considerations has been given more thought into the cost of construction,technology, and equipment setup as well as operational needs.Strategies for cost-effectiveness are prioritized and explored to aid in optimizing equipment purchases and network resources while ensuring that all facilities are future-ready.

Majority of the budget will be directed toward ontruction and essential infrastructure that covers the building materials, structured cabling, and dedicated cooling systems for server rooms.The leftover budget will be set aside for lab equipment, including workstations, network devices, and interactive displays that are sourced from UTM Digital, who can provide reliable products at competitive rates.

Furthermore, the focus on a hybrid approach to data storage by balancing between local servers and cost-effective cloud servers can ensure the scalability without exceeding the budget.Service subscriptions will be selected based on optimal performance and cost-effectiveness, with an emphasis on high-speed connectivity and secure, manageable network infrastructure.

By optimizing resource allocation and equipment procurement, the project is positioned to maximize production value within the allocated budget, supporting the Faculty of Computing's growth with quality, efficient, and future-proof facilities.

Technical Feasibility:

The network infrastructure for the new building we designed for the Faculty of Computing is technically feasible and capable to meet the demands.

The hybrid backup strategy which combines the on-premises and cloud storage is highly feasible for the new FC's building. On-premises solutions like NAS and SAN provide fast data retrieval with low latency, which is suitable for high-usage areas like CNL and ESL. Cloud components like AWS services provide a secure and scalable backup solution for disaster recovery. This approach can minimize the risk of disaster while keeping the operational costs manageable. In addition, the data redundancy across the two locations which is on-premises and cloud ensures the continuity of network service even during major outages.

To ensure the network scalability can accommodate future growth like increasing of student numbers and demand of network resources, a modular architecture which segmented into core, distribution, and access layer will be mainly considered. This architecture enables easy expansion if new areas or services are added. Moreover, hybrid or multi-cloud systems allow dynamic resource allocation which can ensure that the network can handle surges in demand without over-provisioning the physical hardware. Furthermore, applying IPv6 is important to handle the growing number of connected devices which are able to avoid address exhaustion in the future.

To enhance both network security and performance, network segmentation using Virtual LANs (VLANs) will be applied. This technique is used to isolate traffic between the labs, classrooms and student lounge, which is able to minimize network congestion and optimize bandwidth usage. Moreover, VLANs provide a security layer by segmenting the sensitive data within the labs like CNL and ESL from unauthorized access. This network architecture aligns with the FC's needs for a secure, efficient and manageable network infrastructure.

For the building's network architecture, we will combine the Client-Server (CS) for general labs and Peer-to-Peer (P2P) for the CNL which provide a balanced solution for security and functionality. The CS architecture in general labs ensures the centralized control of resources, which makes data management and security easier. The P2P in CNL allows students to simulate direct device-to-device communication and hand-on networking tasks, while the CS will provide administrative control over the sensitive data. This combination of network architecture ensures the scalability, security and efficiency for specialized needs for each facility.

By applying the combination of Category 6a (Cat6a) twisted-pair and fiber optic cabling, it can provide high-speed and stable connectivity to meet FC's performance and scalability requirements. For Cat6a, it supports speeds up to 10 Gbps and bandwidth up to 500 MHz, which will be used for most internal connections for each facility. While the fiber optic cabling which provides higher speed and long-distance capability will be embedded in those critical facilities like CNL and ESL to ensure low-latency network performance where required for specific tasks which require extremely high data throughput.

The Uninterruptible Power Supplies (UPS) and generators will be used for backup power systems to ensure the servers and network operation center (NOC) operate 24/7. The use of UPS will provide immediate protection against the short-term power outages, which is able to protect the data and system. For longer power outages, the generators will ensure the building continues to function without disruption. This is important for the areas like server rooms and NOC which will conduct critical educational activities.

4.0 Appendix - Meeting Minute

DATE/TIME	5 Dec 2024 8:00 pm
LOCATION	M01 KTDI
AGENDA	Task Distribution
Meeting MC	Muhammad Naim bin Abdullah

ATTENDANCE

NAME	TIME	REASON FOR ABSENCE
TAN ZHI MING A23CS0189	2004	
NG YU HIN A23CS0148	2003	
LEE YIN SHEN A23CS0236	2002	
MUHAMMAD NAIM BIN ABDULLAH A23CS0134	2000	

MINUTES

NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task Distribution	Tan Zhi Ming suggested assigning task equally to team members by one small topic and one feasibility, and the meeting minute do by the member who didn't get feasibility task	
2	Choosing Task	Lee Yin Shen decided to take the technical feasibility as his task. Ng Yu Hin decided to take the financial feasibility as his task. Lastly, Tan Zhi Ming decided to take the operational feasibility as the task. Mohammad Naim decided to do Meeting Minute and assist on operational feasibility.	
3	Final Task Distribution	• Mohammad Naim – Meeting Minute and research solution for question • Tan Zhi Ming – Operational Feasibility and research solution for question • Ng Yu Hin – Financial Feasibility and research solution for question • Lee Yin Shen – Technical Feasibility and research solution for question	
5	Meeting ended	21:15	

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